
MATH 493

Algebra

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1 Groups

1.1 Groups and Group Representations

Definition 1.1.1. A group G is a set with a binary operation s.t.

$$(a, b) \rightarrow a \cdot b$$

- Associativity: $(a \cdot b) \cdot c = a(b \cdot c)$
- Identity: $\exists e \in G$ s.t. $a \cdot e = a = e \cdot a$
- Inverse: Given $a \in G$, $\exists b \in G$ s.t. $ab = e = ba$

Example 1.1.2. • Show that the identity is unique

- If $ab = ac$, then $b = c$

- *Proof.* Let e_1, e_2 be identities in G

$$e_1 = e_1 e_2 = e_1$$

□

- *Proof.* $ab = ac$

$$\exists a^{-1} \text{ s.t. } a^{-1}a = e$$

$$a^{-1}(ab) = a^{-1}(ac) = b = c$$

□

Example 1.1.3. • $\{e\}$

- $(\mathbb{Z}, +)$
- $(\mathbb{Q}^\times, \cdot)$
- $\mathbb{Z} / n\mathbb{Z}$ (Cyclic Groups)

$$n \in \mathbb{Z} \text{ if } a, b \in \mathbb{Z} \text{ are congruent with } n \text{ if } n \mid (a - b)$$

Definition 1.1.4. Write $[a]$ as the equivalence class of a .

$$[a] + [b] = [a + b]$$

$$\text{Check if } [a] = [a'] \text{ and } [b] = [b'] \implies [a + b] = [a' + b']$$

Abelian: commutative

- S_3 Permutations of $\{1, 2, 3\}$

$$- (1) = e$$

- (12) = σ
- (13) = $\tau\sigma = \sigma\tau^2$
- (23) = $\sigma\tau$
- (123) = τ
- (132) = τ^2

top first	e	σ	$\sigma\tau$	$\sigma\tau^2$	τ	τ^2
e	e	σ	$\sigma\tau$	$\sigma\tau^2$	τ	τ^2
σ	σ	e	τ	τ^2	$\sigma\tau$	$\sigma\tau^2$
$\sigma\tau$	$\sigma\tau$	τ^2	e	τ	$\sigma\tau^2$	σ
$\sigma\tau^2$	$\sigma\tau^2$	τ	τ^2	e	σ	$\sigma\tau$
τ	τ	$\sigma\tau^2$	σ	$\sigma\tau$	τ^2	e
τ^2	τ^2	$\sigma\tau$	$\sigma\tau^2$	σ	e	τ

This is the same as D_3

$$e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\tau = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 120$$

$$\tau^2 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 240$$

$$\sigma\tau = \begin{bmatrix} -\cos(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 120$$

$$\sigma\tau^2 = \begin{bmatrix} -\cos(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 240$$

1.2 Subgroups

Definition 1.2.1. Let G be a group then $H \leq G$ if

- Contains same Identity
- Closed under group operation
- Closed under taking inverses

Definition 1.2.2. The order of a group, denoted $|G|$ is the cardinality of G

Definition 1.2.3. The order of $a \in G$, the order of a is the smallest $n \in \mathbb{N}$ s.t. $a^n = e$

Theorem 1.2.4 (Lagrange's Theorem). *Let G be a finite group. Let $H \leq G$ then $|H|$ divides $|G|$*

Corollary 1.2.5. *Let $a \in G$ then $\text{ord}(a)$ divides $|G|$*

Definition 1.2.6. Let $a \in G$ and $H \leq G$ then the left coset $aH = \{ah \mid h \in H\}$

Example 1.2.7. $\mathbb{Z}_n$ and let $H = \langle 2 \rangle$

$$a = 0 \text{ "}aH\text{"} = a + H = \langle 2 \rangle$$

$$a = 1 \{1, 3\}$$

$$a = 2 \{0, 2\}$$

$$a = 3 \{1, 3\}$$

Proposition 1.2.8.

1. Any two left cosets are either identical or disjoint
2. G is the disjoint union of the left cosets of H in G

1.3 Normal Subgroups, Quotient groups and the first isomorphism theorem

Definition 1.3.1. A subgroup $H \subset G$ is said to be normal in G if left coset of H in G is also a right coset.

Proposition 1.3.2.

1. H is normal in G
2. $aH = Ha \forall a \in G$
3. $aHa^{-1} = H \forall a \in G$
4. $aHa^{-1} \subseteq H \forall a \in G$
5. $H \subset aHa^{-1} \forall a \in G$

Theorem 1.3.3. Suppose $\varphi : G_1 \rightarrow G_2$ is a group homomorphism then there is a unique isomorphism $\psi : G_1 / \ker \varphi \rightarrow \text{im} \varphi$ s.t. the diagram commutes

$$\begin{array}{ccc}
 G_1 & \xrightarrow{\varphi} & \text{im}(\varphi) \leq G_2 \\
 \downarrow \pi & \nearrow \sim \psi & \\
 G_1 / \ker \varphi & &
 \end{array}$$

Commutes: $\psi(g \ker \varphi) = \varphi(g)$

1.4 Cauchy's Theorem/Conjugation and Conjugacy classes

1.4.1 Cauchy's Theorem Abelian

Theorem 1.4.1. Suppose G is a finite group and $p \mid |G|$ s.t. p is prime the G contains an element of order p .

Proof. Cauchy's Thm for abelian groups

Base Case: $|G| = p$ thus since prove in class G is cyclic thus G has an element of order p

Inductive Step: Assume this works for all groups of order less than n where p divides n .

Suppose $|G| = n$. Let $x \in G$ s.t. $x \neq e$. Let $H = \langle x \rangle \subseteq G$

Case I: $p \mid |H| \implies p \mid \text{ord}(x) \implies \text{ord}(x) = pm$ for $m \in \mathbb{N}$

thus consider $(x^m)^p = e$ thus $\text{ord}(x^m) \mid p \implies x^m$ is order 1 or p it cant be 1 since x is order pm , so it is order p

Case II: $p \nmid |H| \implies H \subsetneq G$ also $H \trianglelefteq G$ since G is Abelian.

Consider $G \rightarrow G/H$

$|G/H| = |G|/|H| < |G|$ since $H \neq \{e\}$ and G/H is abelian

$p \mid |G| \implies p \mid |G/H|$

By induction hypothesis, G/H has an element $\bar{y}$ of order p .

Since the map is surjective there is a $y \in G$ s.t. $\phi(y) = \bar{y}$

$\implies p \mid \text{ord}(y) \implies \text{ord}(y) = pt \implies \text{ord}(y^t) = p$ □

1.4.2 Conjugation and Conjugacy Classes

Definition 1.4.2. Let G be a group and $x, y \in G$

We say x and y are conjugate in G if $\exists g \in G$ s.t. $y = gxg^{-1}$

Proposition 1.4.3. Conjugacy is an equivalence relation

Proof. $x \sim y$ if x is conjugate to y in G

$x \sim x$ $x = exe$

$x \sim y \implies y = gxg^{-1} \implies x = g^{-1}yg \implies y \sim x$

$x \sim y$ and $y \sim z \implies y = gxg^{-1}$ $z = hyh^{-1} \implies z = (hg)x(hg)^{-1} \implies x \sim z$ □

$G = \bigsqcup (\text{Conjugacy classes in } G)$

Example 1.4.4. Suppose G is abelian. $x \sim y \implies y = gxg^{-1} = x \implies$ Conjugacy classes are singletons

$x \sim y \implies \text{ord}(x) = \text{ord}(y)$

Proof. $y = gxg^{-1}$ and $x^m = e$

$$y^m = (gxg^{-1}) \dots (gxg^{-1}) = gx^m g^{-1} = gg^{-1} = e$$

□

thus by the contrapositive if there dont have the same order they cant be in the same class.

Formula for the size of the conjugacy class of an element

Theorem 1.4.5. $|C_x| = |G/Z_G(x)| = \frac{|G|}{|Z_G(x)|}$

First equality for all groups second equality for finite groups

1.5 Sylow Theorems

Theorem 1.5.1. *Let G be a finite group and p be a prime, $p \mid |G|$*

$n = |G| = p^r \times m, p \nmid m$

G contains a subgroup of order p^r .

Theorem 1.5.2. *Any two p -Sylow subgroups of G are conjugate in G .*

Theorem 1.5.3. *The number of p -Sylow subgroups*

1. divides m

2. is congruent to 1 mod p

Example 1.5.4. Classify groups of order 15

n_3 divides 5

$\equiv 1 \pmod{3}$ thus $n_3 = 1$

n_5 divides 3

$\equiv 1 \pmod{5}$ thus $n_5 = 1$

Example 1.5.5. Classify groups of order 21